

Scenario Planning and Sensitivity Analysis — handout

Lesson Forge Pro (owner)

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Lesson plan

Scenario Planning and Sensitivity Analysis

Bloom's levels: Apply, Create

Objectives

- Develop scenario plans to evaluate business outcomes under different assumptions
- Apply sensitivity analysis to test the robustness of financial models
- Create interactive dashboards to visualize scenario results

Prerequisites

- Discounted Cash Flow (DCF) Valuation

Key points

Scenario Planning and Sensitivity Analysis

Session Overview

In this 90-minute session, we will explore the concepts of scenario planning and sensitivity analysis, and how they can be applied to evaluate business outcomes under different assumptions. By the end of this session, you will be able to develop scenario plans, apply sensitivity analysis to test the robustness of financial models, and create interactive dashboards to visualize scenario results.

Timed Agenda

| Time | Topic | Duration |

| --- | --- | --- |

| 0:00 | Introduction and Objectives | 5 minutes |

| 0:05 | Scenario Planning | 20 minutes |

| 0:25 | Energizer/Break | 5 minutes |

| 0:30 | Sensitivity Analysis | 30 minutes |

| 1:00 | Creating Interactive Dashboards | 20 minutes |

| 1:20 | Conclusion and Q&A | 10 minutes |

Introduction and Objectives (0:00 - 0:05)

Welcome to the session on Scenario Planning and Sensitivity Analysis! My name is [Your Name], and I will be your guide through this journey. Before we begin, let's review the objectives of this session:

Develop scenario plans to evaluate business outcomes under different assumptions

Apply sensitivity analysis to test the robustness of financial models

Create interactive dashboards to visualize scenario results

These objectives are aligned with Bloom's levels of Apply and Create, which means we will be applying theoretical concepts to real-world scenarios and creating new content.

Scenario Planning (0:05 - 0:25)

Scenario planning is a technique used to evaluate business outcomes under different assumptions. It's like playing a game of "what if" - what if the market grows at a rate of 10% per annum, what if our competitors launch a new product, what if there's a recession? By creating different scenarios, we can anticipate potential outcomes and develop strategies to mitigate risks or capitalize on opportunities.

Let's consider an analogy. Imagine you're planning a road trip from New York to Los Angeles. You can take different routes, such as the Northern route, the Southern route, or the Central route. Each route has its own set of assumptions, such as traffic, weather, and road conditions. By creating different scenarios, you can anticipate potential delays or hazards and plan accordingly.

Some key concepts in scenario planning include:

Identifying key drivers of change

Developing plausible and relevant scenarios

Evaluating the impact of each scenario on business outcomes

Energizer/Break (0:25 - 0:30)

Let's take a short break to stretch our legs and refresh our minds. Can you think of a time when you had to make a decision under uncertainty? How did you approach it?

Sensitivity Analysis (0:30 - 1:00)

Sensitivity analysis is a technique used to test the robustness of financial models. It's like asking the question - what if our assumptions are wrong? What if the interest rate changes, what if the market grows at a different rate, what if our costs increase? By analyzing the sensitivity of our model to different inputs, we can identify areas of risk and develop strategies to mitigate them.

Let's consider another analogy. Imagine you're building a house, and you're using a blueprint to guide the construction process. The blueprint is like a financial model, and the inputs are like the materials and labor used to build the house. If the price of materials changes, the cost of building the house will also change. By analyzing the sensitivity of the blueprint to different inputs, you can anticipate potential cost overruns and plan accordingly.

Some key concepts in sensitivity analysis include:

Identifying key inputs and outputs

Analyzing the sensitivity of outputs to changes in inputs

Evaluating the robustness of the model to different scenarios

Creating Interactive Dashboards (1:00 - 1:20)

Interactive dashboards are a powerful tool for visualizing scenario results. They allow us to explore different scenarios and see the impact on business outcomes in real-time. It's like having a crystal ball that shows us the potential outcomes of different decisions.

Let's consider an example. Suppose we're evaluating the potential impact of a new product launch on our revenue. We can create an interactive dashboard that shows the revenue forecast under different scenarios, such as a conservative scenario, a moderate scenario, and an aggressive scenario. By exploring the dashboard, we can see the potential risks and opportunities associated with each scenario and make informed decisions.

Conclusion and Q&A (1:20 - 1:30)

That's all the time we have for today's session. I hope you now have a better understanding of scenario planning and sensitivity analysis, and how they can be applied to evaluate business outcomes under different assumptions. Do you have any questions?

Anticipated learner questions:

Q: How do we identify key drivers of change in scenario planning?

A: We can identify key drivers of change by analyzing market trends, competitor activity, and internal factors such as production capacity and supply chain risks.

Q: What is the difference between sensitivity analysis and scenario planning?

A: Sensitivity analysis is a technique used to test the robustness of financial models, while scenario planning is a technique used to evaluate business outcomes under different assumptions. While they are related, they serve different purposes and are used in different contexts.

Q: How do we create interactive dashboards?

A: We can create interactive dashboards using tools such as Excel, Tableau, or Power BI. The key is to identify the key inputs and outputs, and to design a user-friendly interface that allows us to explore different scenarios and see the impact on business outcomes in real-time.